

2023 Ch H2 Q4

Section: Researching Chemistry

Topic: Volumetric Analysis

Question Summary

Describe the experimental procedures, including equipment, used to accurately determine the concentration of a substance by volumetric analysis.

Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q4:

- A standard solution of known concentration is prepared accurately in a volumetric flask.
- A pipette is used to measure a fixed volume of the solution being analysed into a conical flask.
- A few drops of a suitable indicator are added to the conical flask.
- A burette is filled with the standard solution and initial volume is noted.

- The solution is run from the burette into the conical flask with constant swirling until the end point is reached (indicator changes colour).
- The final burette volume is recorded and the titre calculated (final – initial reading).
- The titration is repeated several times until concordant results (within 0.2 cm^3) are obtained.
- The average titre value is used in stoichiometric calculations to determine the unknown concentration.

Final Answer

Volumetric analysis uses a pipette, burette, and standard solution with an indicator. By titrating accurately and obtaining concordant titres, the concentration of the substance can be determined.

Revision Tips

- Always rinse pipette, burette, and conical flask with the correct solutions before use.
- Read burette volumes at eye level to avoid parallax error.
- Only use concordant titres (within 0.2 cm³) for calculations.
- Average titre × concentration (using mole ratio) gives unknown concentration.